Forwarding Test

Consider the following program which implements R = A^2 + B^2

```
LW r1, A
MUL r2, r1, r1      -- A^2
LW r3, B
MUL r4, r3, r3      -- B^2
ADD r5, r2, r4      -- A^2 + B^2
SW r5, R
```

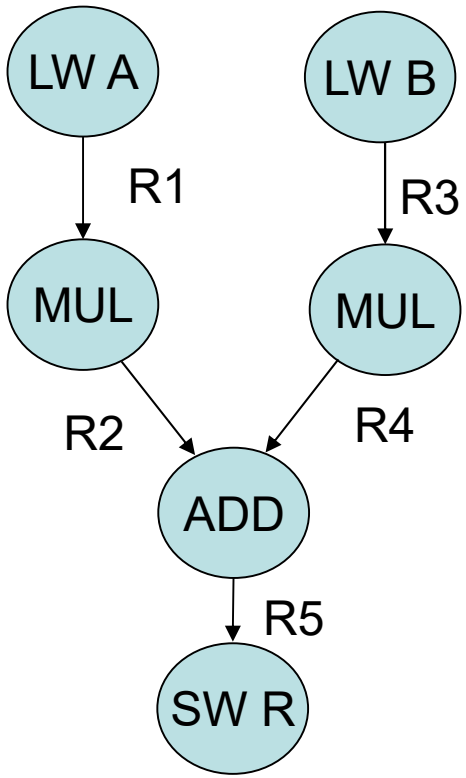
a) Draw its dependency diagram

Forwarding Test

Consider the following program which implements R = A^2 + B^2

```
LW r1, A
MUL r2, r1, r1      -- A^2
LW r3, B
MUL r4, r3, r3      -- B^2
ADD r5, r2, r4      -- A^2 + B^2
SW r5, R
```

a) Draw its dependency diagram



There are only RAW dependencies

Forwarding Test

Consider the following program which implements R = A^2 + B^2

```
LW r1, A
MUL r2, r1, r1      -- A^2
LW r3, B
MUL r4, r3, r3      -- B^2
ADD r5, r2, r4      -- A^2 + B^2
SW r5, R
```

a) Draw its dependency diagram

b) Simulate its execution in a basic 5-stage pipeline <u>without forwarding.</u>

- assume the register file implements "internal forwarding"

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1																				
3	LW R3, B																				
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

MANCHESTER 1824

The University of Manchester

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF																		
3	LW R3, B																				
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	-	EX	MEM	WB												
3	LW R3, B			IF																	
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

MANCHESTER 1824

The University of Manchester

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	-	EX	MEM	WB												
3	LW R3, B			IF	-	-	ID	EX	MEM	WB											
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

MANCHESTER 1824

The University of Manchester

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	-	EX	MEM	WB												
3	LW R3, B			IF	-	-	ID	EX	MEM	WB											
4	MUL R4, R3, R3						IF	ID	-	-	EX	MEM	WB								
5	ADD R5, R2, R4																				
6	SW R5, R																				

MANCHESTER 1824

The University of Manchester

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	-	EX	MEM	WB												
3	LW R3, B			IF	-	-	ID	EX	MEM	WB											
4	MUL R4, R3, R3						IF	ID	-	-	EX	MEM	WB								
5	ADD R5, R2, R4							IF	-	-	ID	-	-	EX	MEM	WB					
6	SW R5, R																				

No Forwarding

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	-	EX	MEM	WB												
3	LW R3, B			IF	-	-	ID	EX	MEM	WB											
4	MUL R4, R3, R3						IF	ID	-	-	EX	MEM	WB								
5	ADD R5, R2, R4							IF	-	-	ID	-	-	EX	MEM	WB					
6	SW R5, R										IF	-	-	ID	-	-	EX	MEM	WB		

With Forwarding

Consider the following program which implements
R = A^2 + B^2

```
LW r1, A
MUL r2, r1, r1      -- A^2
LW r3, B
MUL r4, r3, r3      -- B^2
ADD r5, r2, r4      -- A^2 + B^2
SW r5, R
```

a) Draw its dependency diagram

b) Simulate its execution in a basic 5-stage pipeline without forwarding.

c) Simulate its execution in a basic 5-stage pipeline <u>with forwarding.</u>

- ***Assume the register file implements "internal forwarding"***
- ***Assume MEM-to-EXE forwarding***
- ***Assume WB-to-EXE forwarding***
- ***Assume WB-to-MEM forwarding***

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1																				
3	LW R3, B																				
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1																				
3	LW R3, B																				
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

Data dependency on Load: one bubble + WB-to-EXE bypass

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	EX	MEM	WB													
3	LW R3, B																				
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

Stall because of resource contention on decode logic

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	EX	MEM	WB													
3	LW R3, B			IF	-	ID	EX	MEM	WB												
4	MUL R4, R3, R3																				
5	ADD R5, R2, R4																				
6	SW R5, R																				

Data dependency on Load: one bubble + WB-to-EXE bypass

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	EX	MEM	WB													
3	LW R3, B			IF	-	ID	EX	MEM	WB												
4	MUL R4, R3, R3					IF	ID	-	EX	MEM	WB										
5	ADD R5, R2, R4																				
6	SW R5, R																				

Memory-to-Execute bypass

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	EX	MEM	WB													
3	LW R3, B			IF	-	ID	EX	MEM	WB												
4	MUL R4, R3, R3					IF	ID	-	EX	MEM	WB										
5	ADD R5, R2, R4						IF	-	ID	EX	MEM	WB									
6	SW R5, R																				

A Store is the consumer: Write-Back-to-Memory bypass

With Forwarding

c) Simulate the execution in a 5-stage pipeline with forwarding.

	Instruction	Cycle 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	LW R1, A	IF	ID	EX	MEM	WB															
2	MUL R2, R1, R1		IF	ID	-	EX	MEM	WB													
3	LW R3, B			IF	-	ID	EX	MEM	WB												
4	MUL R4, R3, R3					IF	ID	-	EX	MEM	WB										
5	ADD R5, R2, R4						IF	-	ID	EX	MEM	WB									
6	SW R5, R								IF	ID	EX	MEM	WB								

A Store is the consumer: Write-Back-to-Memory bypass